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Baseline

baseline: 0.975
uniform brightness: 0.975
lighting ramp 0.2: 0.96
lighting ramp 0.4: 0.945
lighting ramp 0.6: 0.895
lighting ramp 0.8: 0.705
rotate 5 deg: 0.97
rotate 10 deg: 0.925
rotate 20 deg: 0.63

Improved

baseline: 0.975
uniform brightness: 0.975
lighting ramp 0.2: 0.98
lighting ramp 0.4: 0.975
lighting ramp 0.6: 0.985
lighting ramp 0.8: 0.975
rotate 5 deg: 0.99
rotate 10 deg: 0.985
rotate 20 deg: 0.98

Method of Improvement

For the lighting changes, I applied normalization to the images. I did this by dividing the image by a Gaussian-blurred copy. Blurring left mostly the lighting changes, and dividing by that left primarily the detailed image, which significantly improved the results. Additionally, 1 was added to the divisor to avoid dividing by 0. Sigma = 8 was chosen due to 2 and 4 producing lower results for the normal photo score, and 16 caused the lighting ramp 0.8 row to produce a lower score. This left sigma = 8 as the best results for all rows.

For the rotation changes, each gallery photo is stored five times, rotated by 20, 10, 0, +10 and +20 degrees, with each copy lighting-normalized after rotation. An image is compared with all of them, and a person is represented by their closest copy. A tilted image therefore finds a stored face at a similar tilt, where more features are more likely to fall in similar positions. The main drawback with this method is that it takes much longer, but the scores are significantly higher.